

RAP 302 Project Execution Methodology

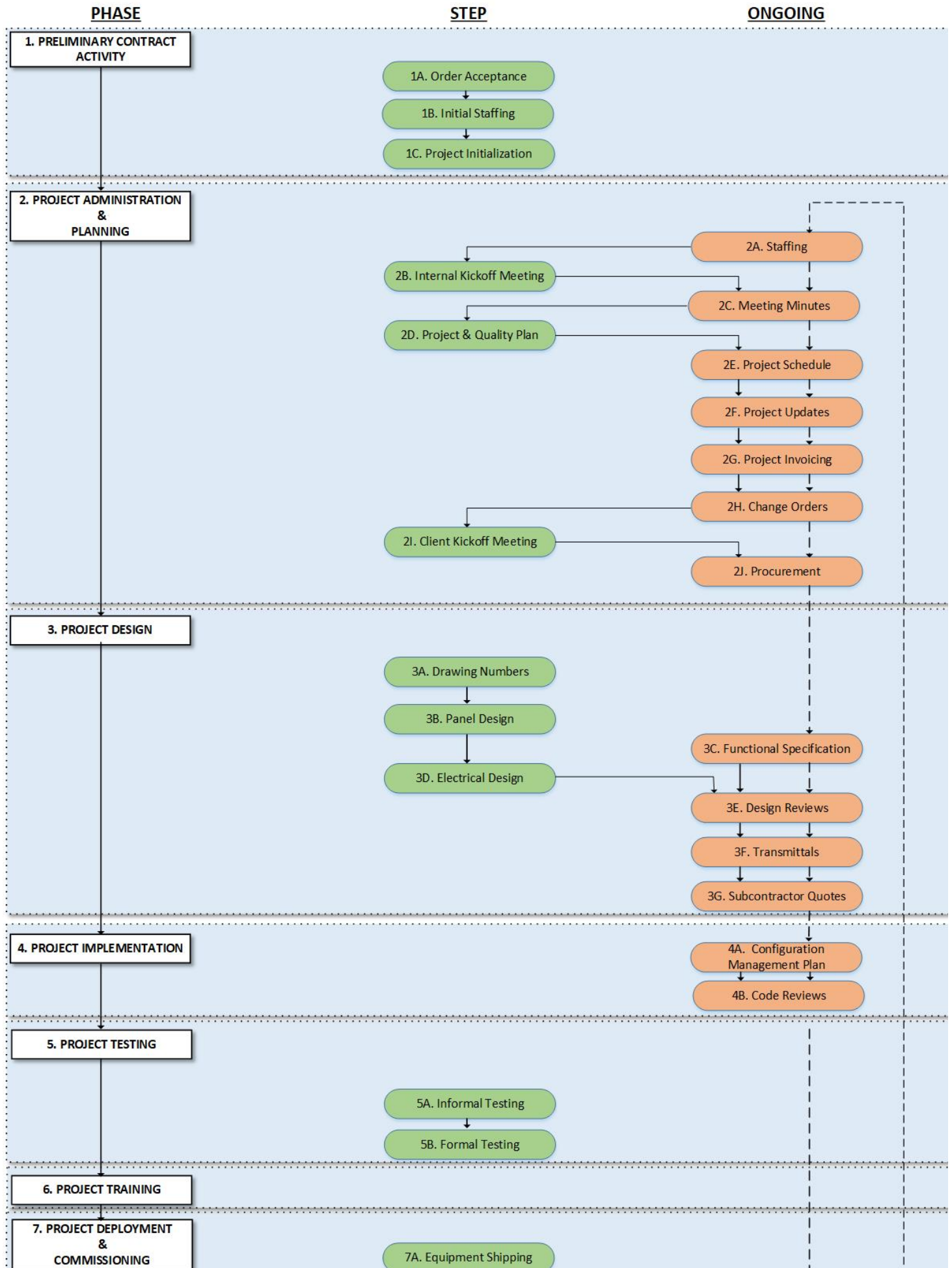
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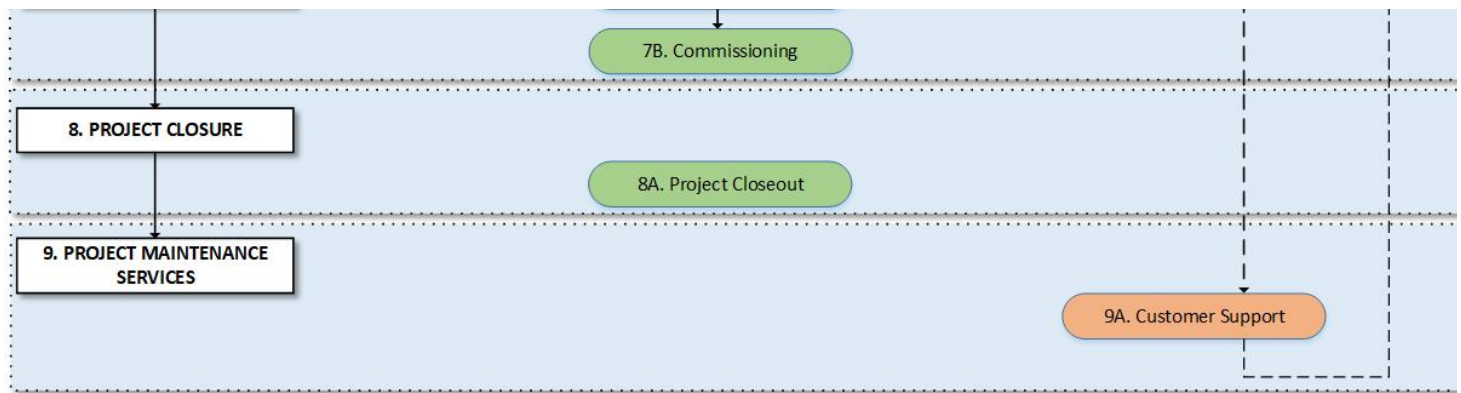
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Approved

Purpose:

This document defines the RoviSys standard project execution methodology. This document defines guidelines that are to be followed to achieve this end result. Not all projects require every item in this document and all project activities may not be completely covered. The execution methodology is illustrated below:





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1. Preliminary Contract Activity

1A. Order Acceptance - Reception of a client's purchase order is the first step in the life of a new project. The acceptance and processing of an order initiates project activities between RoviSys and the client. [Sales Order Acceptance \(RAP 402\)](#)

1B. Initial Staffing - The initial project forecast is a required step in setting up the project within the RoviSys' Staffing tool. The anticipated project schedule along with identification of key project team member roles (and resources if they are available) are documented within the tool to facilitate a proper Internal Kickoff meeting. [Forecasting \(RAP 321\)](#)

1C. Project Initialization Standard -The Project Initialization Standard serves as a guideline when creating projects in GrandView. All billable work including projects, support contracts, service calls, and product sales should be executed through GrandView. [Project Initialization Standard \(RIS 501\)](#)

2. Project Administration & Planning

2A. Staffing - The RoviSys Staffing process is now fully executed per RAP 321 and all project team members required to complete the project should be identified. Due to other project responsibilities, the projects' staffing needs and forecast should be reviewed on a regular basis to ensure proper resource availability and loading. [Forecasting \(RAP 321\)](#)

2B. Internal Kick-Off Meeting - The Internal Kickoff Meeting serves as a formal transfer of project information from the RoviSys Sales team to the RoviSys Project team. Any project risks that were identified by the Sales Person(s) should be captured and reviewed with the Project Team during this meeting. [Internal Kickoff Meeting \(RIS 503\)](#)

2C. Meeting Minutes - Meeting minutes are the instant written record of a meeting. All RoviSys project meetings (both internal and external) are required to have documented meeting minutes, including PAIL items if applicable, which are distributed to all who attended the meeting (and others as seen fit). [Meeting Minutes \(RIS 518\)](#)

2D. Project & Quality Plan - The RoviSys Quality and Project Plan incorporates both the references associated with all of the planning requirements for a project along with how RoviSys' project team will ensure that the quality of the project meets and/or exceeds the client's expectations. [Quality & Project Plan \(RIS 541\)](#)

2E. Project Schedule - Establishing and working to a realistic schedule is intended to provide the greatest opportunity for success to a project. Schedule management includes the processes required to ensure timely completion of the project by working to the schedule, evaluating status and communicating regularly with all project members. [Project Schedule \(RIS 505\)](#)

2F. Project Updates - Project Update meetings keep the project team informed of the status of the project. They summarize work completed, identify work to be completed, identify any technical issues requiring resolution, review risk mitigation and review any changes to project schedule. If Project Action Item List (PAIL) is being utilized on the project, all PAIL items should be reviewed and updated accordingly. [Project Updates \(RIS 523\)](#)

2G. Project Invoicing - Invoices are issued on a project basis and are the method for collecting payment from our customers for work that has been completed. Invoices are used to collect payment for engineering services, OPE and T&L expenses that were expended during the billing period. [Project Invoicing Policy \(RAP 202\)](#)

2H. Change Orders - Any changes to RoviSys' scope of work requiring additional engineering services, software or hardware should be documented and discussed with the client prior to execution of the additional services or prior to ordering any necessary additional software or hardware. Change order procedures should be discussed with the client as part of the Client Kickoff meeting. [Change Orders \(RAP 403\)](#)

2I. Client Kickoff Meeting - The Client Kickoff meeting is the first time that the entire project team (RoviSys, client, sub-contractors) gets together to review the project. The goal is to review items such as scope, schedule, risks, communication channels and finances. [Client Kickoff Meeting \(RIS 504\)](#)

2J. Procurement - Procurement is the process by which any hardware and/or software that are required to be purchased by RoviSys as part of a client project is conducted. Administratively, quotes need to be obtained (or verified) and vendor contacts need to be established. This is a re-occurring task as many items cannot be identified until completion of a project's Design phase. Often, additional items are required in order to complete commissioning and startup activities. [Purchase Orders \(RAP 307\)](#)

3. Project Design

3A. Drawing Numbers - For those projects that require CAD-based drawings, it is important to discuss with the client whether they have specific drawing number requirements or if they are agreeable to using RoviSys' own standards. [Drawing Numbers \(RIS 517\)](#)

3B. Panel Design - A successful panel build is the result of following a structured format where all steps must be adhered to in order to design/build a quality cabinet product for the client. [Panel Build Guidelines \(RIS 522\)](#)

3C. Functional Specification - The Functional Specification document defines the overall requirements for the system and is the basis of design for the project's implementation phase. [Functional Specification \(RIS 507\)](#)

3D. Electrical Design - If a project includes electrical design services, it is important that the actual design follows strict guidelines to ensure conformity to applicable regulations and results in a system that is safe and does not present any degradation in overall control system performance. [Engineering & Design \(RIS 516\)](#)

3E. Design Reviews - The purpose of the Design Review is to establish that the project deliverables are in accordance with system requirements and standards, identify any reuse candidates and issues prior to end user review, and propose remedial actions where required. Design Reviews shall be performed by a project team member, other than the author, prior to transmittal. [Design Reviews \(RIS 526\)](#)

3F. Transmittals - The purpose of transmittals is to formally document the transfer of documentation and/or information to the client. Transmittals can consist of drawing packages, documents, software configuration and other applicable sources of information. [Transmittal Standard \(RIS 519\)](#)

3G. Subcontractor Quotes - When a subcontractor or vendor is involved with a project it is important to follow RoviSys' standard procedure for the approval and administration of agreements between RoviSys and our subcontractors/vendors. These agreements, which also apply to our clients, include Non-Disclosures, Master Services, Multi-Site Pricing, Alliance, Licensing and Partnerships. [Customer & Vendor Agreement Process \(RAP 404\)](#)

4. Project Implementation

4A. Configuration Management Plan - Configuration Management establishes the systematic control and recording of the development and changes that are made to the hardware, software and documentation so that a system or a subsystem component maintains its integrity throughout the system development life-cycle, including internal assets, re-use items, as well as project deliverables. [Configuration Management Plan \(RIS 537\)](#)

4B. Code Reviews - Code reviews provide guidance for developers and helps promote quality code for projects. The core mission of a code review is to ensure that Developers/Engineers are developing the right solutions on a sustainable and supportable foundation. [Code Reviews Standard \(RIS 525\)](#)

4C. Third Party Licenses - Third-party libraries used within a project must be licensed with a license that RoviSys can use. Refer to the Approved third-party Licenses list for details. All third-party libraries used on a project must be documented along with their license names in a document that is transmitted to the customer. [Approved third-party licenses](#)

5. Project Testing

5A. Informal Testing - Informal testing should occur prior to customer acceptance testing to ensure that the system is accurately designed to reflect the intent of the project. [Pre-FAT System Checkout \(RIS 538\)](#)

5B. Formal Testing - Formal testing of systems is a critical step in ensuring that the design and functionality of the systems conform to the client's needs and expectations. Testing procedures should reflect the requirements spelled out in the Functional Specification. [System Testing \(RIS 543\)](#)

6. Project Training

RoviSys typically provides "hands-on" informal training as part of our commissioning and startup activities. Some projects require a more formal "classroom" style of training and it is important to develop training materials that accurately reflect the system being provided as well as how it interacts with the process and application.

7. Project Deployment & Commissioning

7A. Equipment Shipping - Customer owned equipment in RoviSys possession will be shipped to the customer site or third party vendor prior to commissioning. [Shipping and Returns \(RAP 312\)](#)

7B. Commissioning - Commissioning includes onsite activities to verify functionality of the system through testing and verification. A Site Acceptance Test (SAT) is often part of the onsite activities required of a RoviSys engineering team. The SAT Test Plan is utilized to enable a formal onsite testing procedure that accurately reflects the original intent and scope of the project as documented within the Functional Specification. [Commissioning \(RIS 521\)](#), [SAT Test Plan \(RIS 540\)](#)

8. Project Closure

8A. Project Closeout - Project closure is essential in ensuring that all of the project's proposed deliverables have been provided and that all RoviSys' internal project closure procedures have been completed. [Project Closeout \(RAP 308\)](#)

9. Project Maintenance Services

9A. Customer Support - The RoviSys Customer Support policy provides the methods and tools for assigning and tracking support requests from customers who are operating systems designed or modified by RoviSys. [Customer Support \(RAP 314\)](#)

Page History

Revision	Action/Comment	Date	User
r31 (this)	Approve	20-Jun-2023 5:53 UTC	Hayley Williams
r31 (this)	Create/Edit <i>Removed reference to obsolete RIS 527</i>	20-Jun-2023 12:21 UTC	Beverly Kiglics
r30	Approve	21-Mar-2023 5:33 UTC	Beverly Kiglics
r30	Create/Edit <i>PECII-Periodic Review</i>	21-Mar-2023 4:46 UTC	Dave Hermann
r29	Approve	14-Dec-2021 5:42 UTC	Joshua Hilewick
r29	Create/Edit <i>PECII-Changed link in 3D to RIS 516</i>	14-Dec-2021 4:54 UTC	Dave Hermann
r28	Approve	12-Mar-2021 7:29 UTC	Colin Waid
r28	Create/Edit <i>Updated Visio drawing section 5</i>	12-Mar-2021 7:13 UTC	Dave Hermann
r27	Approve	21-Aug-2020 3:35 UTC	Jeff Marshall
r27	Create/Edit <i>Updated Section 5</i>	12-Mar-2020 5:19 UTC	Kit Phillips
r26	Approve	11-Sep-2019 1:24 UTC	Jeff Marshall
r26	Create/Edit <i>adding 4C - third party licenses</i>	27-Dec-2018 1:41 UTC	Patrick Mechenbier
r25	Create/Edit <i>adding 4C - third party licenses</i>	27-Dec-2018 1:41 UTC	Patrick Mechenbier
r24	Approve	13-Dec-2018 5:32 UTC	Jeff Marshall
r24	Create/Edit <i>Formatting Changes</i>	28-Nov-2018 8:38 UTC	Martin Gonzalez
r23	Approve	6-Jun-2018 5:21 UTC	John Turk
r23	Create/Edit <i>Major update as a result of input ove...</i>	30-May-2018 6:49 UTC	Dave Smrdel
r22	Approve	15-Jan-2018 4:38 UTC	John Turk
r22	Create/Edit <i>Update per President's request and to...</i>	12-Jan-2018 6:55 UTC	Dave Smrdel
r21	Approve	4-Oct-2017 4:10 UTC	Jeff Marshall
r21	Create/Edit <i>Added link to staffing process (discu...</i>	17-Aug-2017 4:49 UTC	MaryJo Chelmins
r20	Approve	24-Aug-2015 3:12 UTC	John Turk
r20	Create/Edit <i>Software Congress Proposal</i>	4-Jun-2015 1:54 UTC	Chris Conry
r19	Approve	15-Jan-2015 5:06 UTC	Jeff Marshall
r19	Create/Edit <i>PECII Review - Minor edits</i>	15-Jan-2015 5:05 UTC	Dave Hermann
r18	Approve	4-Dec-2014 9:24 UTC	Jeff Marshall

Revision	Action/Comment	Date	User
r18	Create/Edit <i>PECII Review - Minor edits</i>	4-Dec-2014 6:08 UTC	Dave Hermann
r17	Approve	11-Jan-2013 12:35 UTC	Jeff Marshall
r17	Create/Edit <i>Fixed tags</i>	10-Jan-2013 12:18 UTC	Dave Hermann
r16	Approve	17-Aug-2012 1:20 UTC	Pete Madonia
r15	Approve	17-Aug-2012 1:19 UTC	Pete Madonia
r16	Create/Edit <i>Revised some grammatical issues</i>	30-May-2012 3:00 UTC	Ruoth, Joel
r15	Create/Edit <i>Added link to Risk Standard to Intern...</i>	4-May-2012 4:12 UTC	Bryan DeBois
r14	Approve	29-Mar-2011 9:38 UTC	Dick Ciammaichella
r14	Create/Edit <i>Removed too-specific tag</i>	10-Mar-2011 3:57 UTC	Bryan DeBois
r13	Create/Edit <i>Added [302] as tag</i>	7-Mar-2011 6:17 UTC	Derek Drayer
r12	Create/Edit <i>Updated Title to Project "Execution"...</i>	7-Mar-2011 4:50 UTC	Jeff Marshall
r11	Approve	7-Mar-2011 2:15 UTC	John Turk
r11	Unapprove	7-Mar-2011 2:15 UTC	John Turk
r11	Approve	3-Mar-2011 9:08 UTC	Tom Grulkowski
r11	Create/Edit <i>removed rap-candidate tag</i>	15-Feb-2011 3:40 UTC	Haren, Michael
r10	Approve	2-Dec-2010 12:55 UTC	Haren, Michael
r10	Unapprove	2-Dec-2010 12:55 UTC	Haren, Michael
r10	Approve	6-Oct-2010 11:53 UTC	Jeff Marshall
r10	Create/Edit <i>links to raps and standards, mild for...</i>	6-Oct-2010 7:09 UTC	Xander Dumaine
r9	Create/Edit <i>links to raps and standards, mild for...</i>	6-Oct-2010 6:59 UTC	Xander Dumaine
r8	Create/Edit <i>Completely revised from word doc</i>	5-Oct-2010 1:39 UTC	Xander Dumaine
r7	Approve	1-Oct-2010 1:51 UTC	John Turk
r7	Create/Edit <i>Imported rev. g of RAP into RoviPedia</i>	2-Sep-2010 6:40 UTC	Tom Grulkowski
r6	Create/Edit <i>Guess it did</i>	1-Sep-2010 4:14 UTC	Bryan DeBois
r5	Create/Edit <i>Won't work</i>	1-Sep-2010 4:14 UTC	Bryan DeBois
r4	Create/Edit <i>image fix</i>	27-Aug-2010 3:42 UTC	Xander Dumaine
r3	Create/Edit <i>formatting</i>	27-Aug-2010 3:25 UTC	Xander Dumaine

Revision	Action/Comment	Date	User
r2	Create/Edit <i>image</i>	27-Aug-2010 3:08 UTC	Xander Dumaine
r1	Create/Edit <i>Initial revision</i>	27-Aug-2010 3:05 UTC	Xander Dumaine

Note: This is a snapshot of a living document. Please check online for the latest revision.

Edited: 20-Jun-2023 12:21 UTC **Printed:** 20-Oct-2023 6:52 UTC

Document Number: 137 **Revision:** r31